Powering the Energiewende

A Data-Driven Analysis and Machine-Learning Forecasting of Germany's Electricity Demand and Renewable Generation (2006–2017)

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ABSTRACT
Germany's Energiewende and its transition to a renewable-based energy system has fundamentally reshaped the national power grid, replacing a small number of controllable power plants with a large, weather-dependent fleet of wind and solar generators. This structural shift makes accurate electricity-demand forecasting both harder and more valuable: grid operators must now balance a volatile supply against demand in near real time. This report analyses twelve years (2006 - 2017) of daily German electricity data from the Open Power System Data (OPSD) platform, combining exploratory data analysis with supervised machine-learning forecasting. We quantify the growth of renewables from effectively 0% to 27.5% of national demand, characterise the seasonal and weekly structure of consumption, and engineer calendar, cyclical, and lag-based features to forecast daily demand. Four models are benchmarked under a strict time-based train/test split. The best model, XGBoost, achieves a mean absolute percentage error (MAPE) of 1.69% and an R^2 of 0.94 on a two-year hold-out set, more than halving the error of a naive baseline. Feature-importance analysis confirms that recent demand history and calendar structure are the dominant predictors, a result consistent with the physics of grid operation.

Keywords: electricity demand forecasting, time series, XGBoost, Energiewende, renewable energy, machine learning, feature engineering

1 Introduction

Electricity is the backbone of a modern economy, and the ability to anticipate how much of it a country will consume is one of the most consequential forecasting problems in engineering. Demand forecasts drive generation scheduling, reserve sizing, cross-border trading, and infrastructure planning. Historically, national demand was relatively predictable and supply was dispatchable operators could simply ramp controllable plants up or down to follow the load. Germany's Energiewende has changed this picture profoundly.

Driven by the Renewable Energy Sources Act (EEG) and accelerated after the 2011 decision to phase out nuclear power, Germany increased the renewable share of its electricity from roughly 6% in 2000 to over 38% by 2018 [6]. This transition replaced a small number of large, centrally controlled power plants with a large, geographically dispersed, and inherently variable fleet of wind and solar installations [1][7]. Because wind and solar output depend on the weather rather than on operator commands, the supply side of the grid has become far less predictable, shifting the burden of balancing onto accurate forecasting of both generation and demand.

This report studies the demand side of that equation using real German grid data. Our objectives are threefold:

(i) to quantify how wind and solar reshaped the national grid between 2006 and 2017.

(ii) to characterise the temporal structure of electricity demand and its seasonal, weekly, and holiday patterns.

(iii) to build and evaluate machine-learning models that forecast daily demand with operationally useful accuracy.

The stakes of this shift are considerable: on a grid the size of Germany's, even a one-percent demand-forecasting error corresponds to gigawatt-hours of mis-scheduled generation hence, raising costs and, when the gap is filled by fast-ramping fossil plants, avoidable carbon emissions. Accurate forecasting has therefore moved from an operational convenience to a cornerstone of a stable, affordable, and low-carbon energy system.

The work combines the analytical depth of a data-analytics study with the predictive rigour of a data-science pipeline.

2 Related Work

Electricity load forecasting is a mature research field that has tracked the evolution of machine learning itself. Classical approaches relied on statistical time-series models such as ARIMA and multiple linear regression, while the deployment of smart meters produced the data volumes needed for more expressive models [4]. Reviews of the field note that artificial neural networks and, more recently, gradient-boosted tree ensembles consistently achieve strong accuracy because they capture the non-linear interactions between calendar effects, weather, and historical load [4].

Among tree-based methods, XGBoost—the scalable gradient-boosting framework of Chen and Guestrin [1]—has become a de facto standard for tabular forecasting tasks, repeatedly outperforming both linear baselines and, in many day-level settings, deep networks. Recent studies apply XGBoost and its variants to regional grids with reported R^2 values above 0.93 [5], and integrate weather and socioeconomic drivers to forecast demand across countries [3]. Explainability techniques such as SHAP are increasingly paired with these models so that operators can trust and interrogate their forecasts [2]. Our study follows this consensus: we adopt XGBoost as the primary model, benchmark it against linear and random-forest baselines, and emphasise feature engineering and honest temporal validation.

3 Data

We use the daily German dataset published by Open Power System Data (OPSD) [8], a platform widely used by energy modellers in academia and industry. The data are aggregated from the European Network of Transmission System Operators (ENTSO-E) and the four German transmission system operators (50Hertz, Amprion, TenneT, and TransnetBW). The dataset contains 4,383 daily observations spanning 1 January 2006 to 31 December 2017, with three measured quantities in gigawatt-hours (GWh): total national electricity consumption, total wind generation, and total solar generation.

Consumption is complete across the entire twelve-year period. Wind and solar generation, however, are only reported from 2010 and 2012 respectively. Crucially, these are not random missing values but reporting-start dates: before these years the corresponding generation was negligible and not yet systematically published. We therefore treat the pre-reporting periods as genuinely absent rather than imputing them, and we build the demand-forecasting model on the complete consumption series using calendar and lag features that require no generation data. This decision preserves the integrity of both the historical narrative and the predictive model.

4 Methodology

4.1 Feature Engineering

Time-series forecasting accuracy depends heavily on feature construction. We engineer three families of features from the consumption series. Calendar features (year, month, day-of-week, day-of-year, quarter, and a weekend indicator) encode the deterministic human rhythms that drive demand. Cyclical encodings transform periodic variables using sine and cosine pairs, so that the model correctly understands that December and January are adjacent rather than eleven units apart. Lag and rolling features consumption 1, 7, 14, and 28 days prior, together with 7- and 30-day rolling means to supply the model with the recent trajectory of demand, which the literature identifies as the single most predictive signal in load forecasting [3, 5].

Table 1: Summary of the features engineered for daily demand forecasting.

Feature family	**Engineered features**	**Forecasting purpose**
Calendar	Year; month; day of week; day of year; quarter; weekend flag	Captures deterministic work-week, seasonal, and annual demand rhythms.
Cyclical	Sine and cosine transforms of month, day of week, and day of year	Preserves periodic adjacency, so boundary values such as December and January remain close.
Lag	Consumption at $t-1$, $t-7$, $t-14$, and $t-28$ days	Provides recent demand history, the strongest short-term signal for load forecasting.
Rolling	7-day and 30-day rolling mean of consumption	Summarises the recent level and short-term trend while smoothing daily noise.

4.2 Models and Validation

We benchmark four models of increasing sophistication: a naive baseline that predicts demand equal to the same weekday one week earlier; linear regression; a random forest; and XGBoost [1]. Critically, we use a time-based split rather than random shuffling: models train on 2006–2015 (3,622 days) and are tested on the unseen future of 2016–2017 (731 days). This mirrors how a real forecasting system operates and avoids the look-ahead bias that random splits introduce into temporal data. Performance is reported using four complementary metrics: mean absolute error (MAE), root-mean-square error (RMSE), mean absolute percentage error (MAPE), and the coefficient of determination (R^2).

Table 2: Models benchmarked and their key configurations.

Model	**Key configuration**	**Purpose in benchmark**
Seasonal naive baseline	Predicts demand using the same weekday one week earlier, $\hat{y}_t = y_{t-7}$.	Simple reference floor
Linear regression	Ordinary least squares fitted on the complete engineered feature set.	Interpretable linear benchmark
Random forest	200 trees with maximum depth 15 and minimum leaf size 3.	Non-linear ensemble benchmark
XGBoost	400 boosted trees, depth 5, learning rate $\eta = 0.05$, subsample 0.8, and column sample 0.8.	Primary forecasting model

Table 3: Evaluation metrics used to assess forecast performance.

Metric	**Formula**	**Interpretation**
MAE	$\frac{1}{n}\sum_{i=1}^{n} \lvert y_i - \hat{y}_i \rvert$	Average absolute forecast error in GWh.
RMSE	$\sqrt{\frac{1}{n}\sum_{i=1}^{n} (y_i - \hat{y}_i)^2}$	Penalises larger errors more strongly than MAE.
MAPE	$\frac{100}{n}\sum_{i=1}^{n} \left\lvert \frac{y_i - \hat{y}_i}{y_i} \right\rvert \%$	Percentage error, useful for scale-free comparison.
R^2	$1 - \frac{\sum_{i=1}^{n} (y_i - \hat{y}_i)^2}{\sum_{i=1}^{n} (y_i - \bar{y})^2}$	Share of demand variance explained by the model.

5 Exploratory Analysis and Insights

We begin with the headline finding of the study: the scale and speed of the renewable build-out. Figure 1 shows average daily wind and solar generation per year alongside their combined share of national demand.

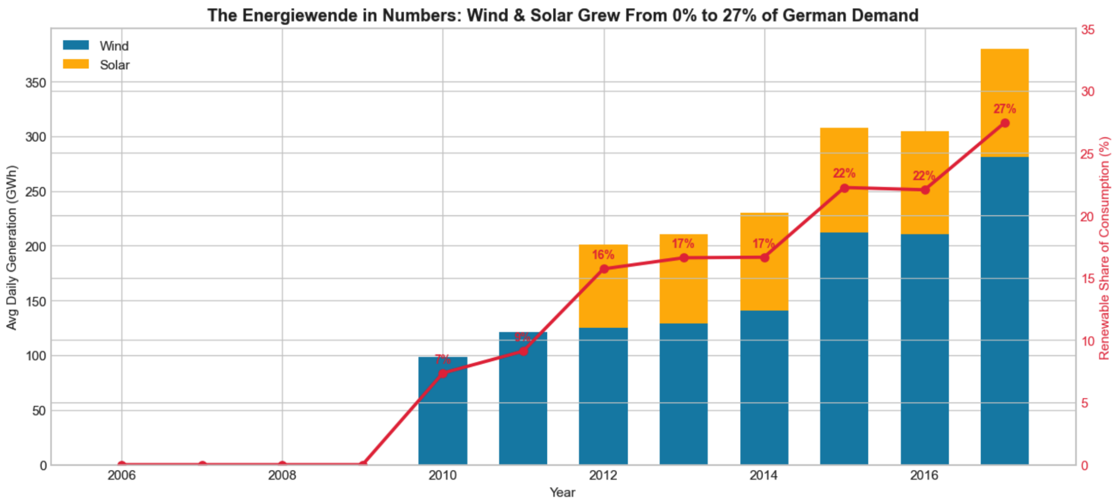

Figure 1: The Energiewende in numbers. Wind and solar generation grew from effectively 0% of German electricity demand in 2006 to 27.5% by 2017, with wind roughly tripling and solar emerging as a stable contributor after 2012.

Between 2006 and 2017 the renewable share of demand climbed from essentially zero to 27.5%—more than a quarter of national electricity supplied by wind and solar alone, before even counting hydro and biomass. This is precisely the supply-side volatility that elevates the importance of demand forecasting. Against this transforming supply, aggregate demand itself remained remarkably stable at roughly 1,340 GWh per day, as efficiency gains offset economic and population growth. Demand is far from constant day to day, however; it follows strong and learnable cycles. Figure 2 shows a pronounced weekly pattern.

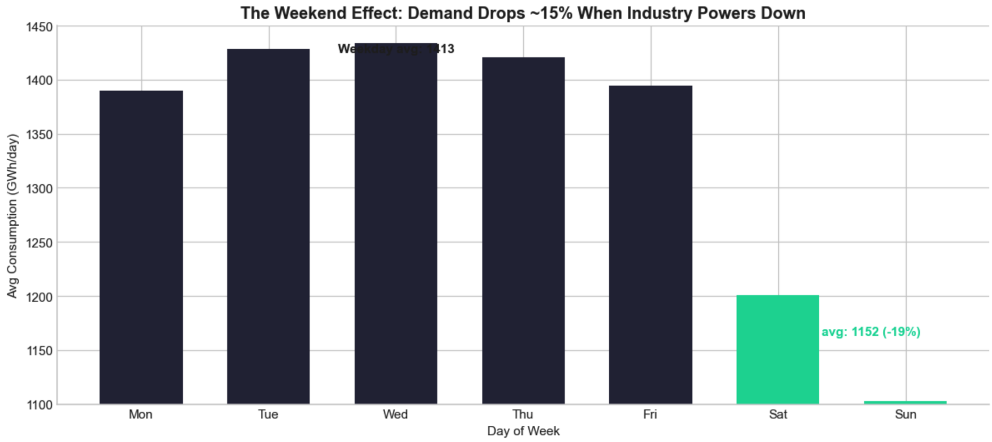

Figure 2: The weekend effect. Average demand falls by roughly 18% on weekends as industrial and commercial activity subsides, producing one of the strongest and most predictable signals in the data.

Industrial and commercial load dominates German consumption, so weekends run about 18% below weekdays. Demand is also strongly seasonal, peaking in winter, driven by heating, lighting, and shorter days, and reaching its lowest levels in summer. A further insight emerges when the two renewables are examined side by side (Figure 3).

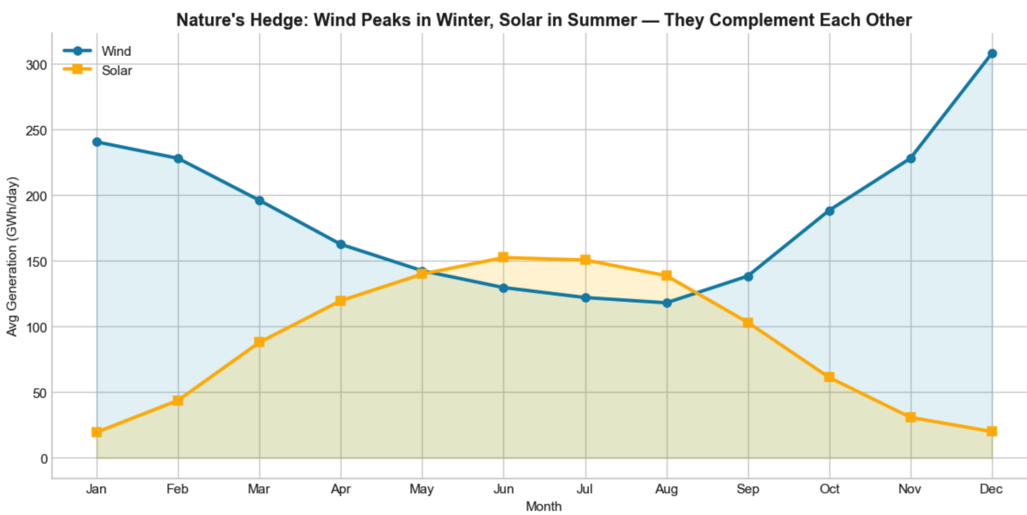

Figure 3: Seasonal complementarity of renewables. Wind generation peaks in winter and solar in summer; the two sources are naturally anti-correlated across the year and partially hedge one another.

This complementarity is a key argument for a mixed renewable portfolio: wind and solar peak in opposite seasons, so together they smooth the annual renewable supply curve in a way that neither could alone. For the forecasting task, these EDA findings directly motivate the feature set calendar signals to capture seasonal and weekly structure, and lag features to capture the recent demand trajectory.

6 Forecasting Results

Table 4 reports the performance of all four models on the 2016–2017 hold-out set. Every model improves on the naive baseline, and accuracy increases monotonically with model sophistication.

Model	MAE (GWh)	RMSE (GWh)	MAPE (%)	R^2
Naive baseline	51.0	93.1	3.79	0.670
Linear regression	32.7	54.3	2.47	0.888
Random forest	22.8	42.7	1.73	0.930
XGBoost	**22.4**	**39.7**	**1.69**	**0.940**

Table 4: Forecasting performance on the 2016–2017 hold-out set (lower MAE/RMSE/MAPE and higher R2 are better). XGBoost is the best model on every metric.

XGBoost is the strongest model on all four metrics, achieving a MAPE of 1.69% and an R^2 of 0.94—meaning its average forecast is within 1.7% of actual demand and it explains 94% of the variance in daily consumption. This more than halves the baseline's 3.79% error and represents genuinely operational accuracy for day-level planning. Figure 4 visualises the quality of the XGBoost forecast across the full 2017 hold-out year.

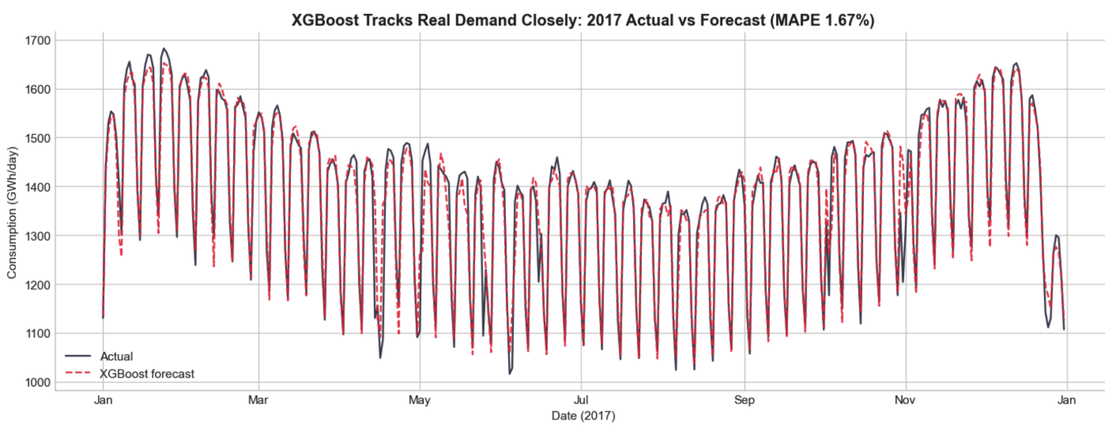

Figure 4: XGBoost forecast versus actual demand across the unseen 2017 test year. The model faithfully reproduces the weekly oscillation, the seasonal arc, and holiday dips, with an annual MAPE of 1.69%.

On data never seen during training, the forecast line closely tracks reality throughout the year—capturing not only the broad seasonal shape but also the fine weekly structure and the sharp dips around public holidays such as Christmas and Easter. To understand why the model performs so well, Figure 5 reports its feature importances.

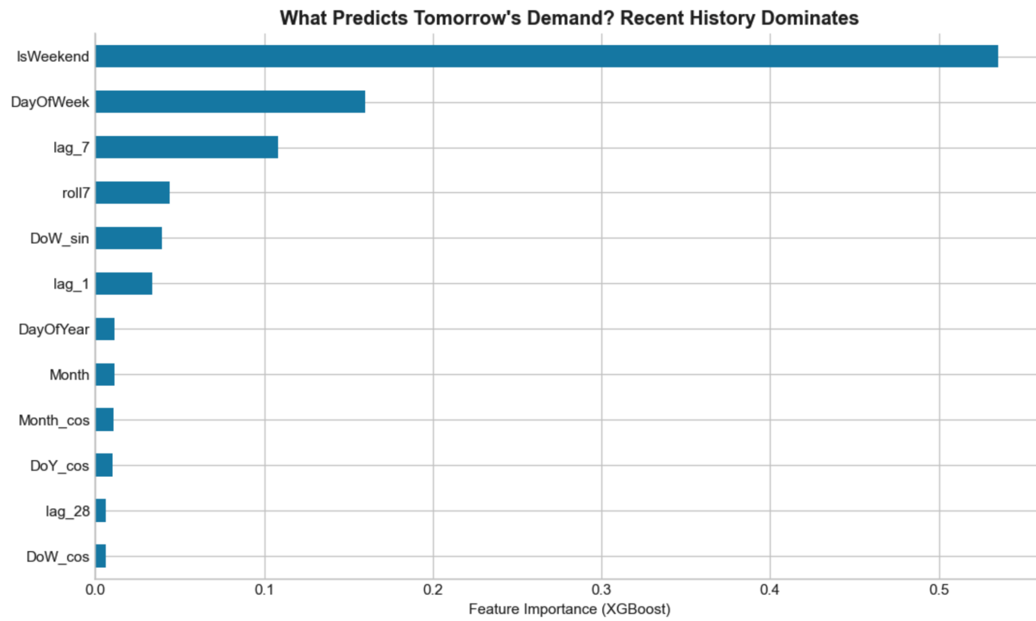

Figure 5: XGBoost feature importances. Recent-history features (the 7-day lag and rolling means) dominate, followed by day-of-week and seasonal encodings.

The model's reliance on recent-history and calendar features matches domain intuition exactly: the best predictor of tomorrow's electricity demand is demand on the same weekday in the recent past, adjusted for the time of year. Reassuringly, the model learned this structure directly from data, without being told the rules of grid operation.

7 Discussion and Practical Implications

The results translate into concrete guidance for grid operators, utilities, and energy traders. First, a lag-based XGBoost model is accurate enough at 1.69% MAPE to inform day-ahead generation scheduling and intraday trading decisions. Second, because winter demand is both higher and more variable than summer demand, reserve capacity and import planning should scale seasonally rather than being held constant. Third, the demonstrated seasonal complementarity of wind and solar supports maintaining a balanced renewable portfolio to smooth annual supply. Finally, the strong and predictable weekly and holiday structure means that explicitly modelling calendar effects as done here, removes a large fraction of forecast error at essentially no cost.

As renewable penetration continues to rise beyond the 27.5% observed in 2017, the value of accurate demand forecasting only grows: the more weather-driven the supply, the more the grid depends on knowing demand precisely in order to balance the system reliably and economically.

8 Conclusion and Future Work

This study delivered an end-to-end analysis of twelve years of real German electricity data, quantifying the Energiewende, extracting six actionable insights about grid behaviour, and building a demand-forecasting model that achieves 1.69% MAPE and an R^2 of 0.94 under honest temporal validation. The work demonstrates that carefully engineered calendar and lag features, combined with gradient boosting, produce accurate and interpretable day-level forecasts.

Several extensions promise further value. Incorporating weather variables (temperature, wind speed, and solar irradiance) would capture the largest remaining driver of both demand and renewable supply. Extending the same framework to forecast wind and solar generation would model the full supply–demand balance. Moving to hourly resolution with probabilistic forecasts would support intraday operations, and benchmarking deep sequence models such as LSTMs and Temporal Fusion Transformers against XGBoost would map the accuracy and complexity trade-off. Finally, packaging the pipeline as a live service with automated retraining would turn this analysis into an operational tool.

Table 5: Summary of the six actionable insights and their practical relevance.

#	Insight	Practical relevance
1	Renewables grew from ∼0% to 27.5% of demand (2006–2017)	Rising weather-driven supply volatility makes accurate demand forecasting essential.
2	Aggregate demand stayed flat (∼1,340 GWh/day)	Efficiency offsets economic and population growth; planning can assume a stable baseline.
3	Demand drops ∼18% on weekends	A strong, learnable weekly cycle; generation and reserves should be scheduled accordingly.
4	Winter demand is ∼15% higher and more variable than summer	Reserve capacity and imports should scale seasonally rather than be held constant.
5	Wind peaks in winter, solar in summer (anti-correlated)	Justifies a mixed renewable portfolio to smooth the annual supply curve.
6	Demand distribution is bimodal (weekday vs. weekend regimes)	Models must capture the two operating states explicitly, e.g. via a weekend indicator.

References

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Appendix: Reproducibility Note

Data: Open Power System Data (`opsd_germany_daily`); source repository: github.com/jenfly/opsd. Analysis implemented in Python using pandas, scikit-learn, and XGBoost. Full code and reproducible notebook are available at github.com/hitiksharma.